

Name: _____

St George Girls High School

Trial Higher School Certificate Examination

2015



Mathematics

Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using blue or black pen.
- Write your student number on each booklet.
- Board-approved calculators may be used.
- A table of standard integrals is provided at the back of this paper.
- The mark allocated for each question is listed at the side of the question.
- Marks may be deducted for careless or poorly presented work.

Total Marks – 70

Section I – Pages 2 – 5

10 marks

- Attempt Questions 1 – 10.
- Allow about 15 minutes for this section.
- Answer on the sheet provided.

Section II – Pages 6 – 11

60 marks

- Attempt Questions 11 – 14.
- Allow about 1 hour 45 minutes for this section.
- Begin each question in a new booklet.
- Show all necessary working in Questions 11 – 14.

Students are advised that this is a Trial Examination only and does not necessarily reflect the content or format of the Higher School Certificate Examination.

Section I

10 marks

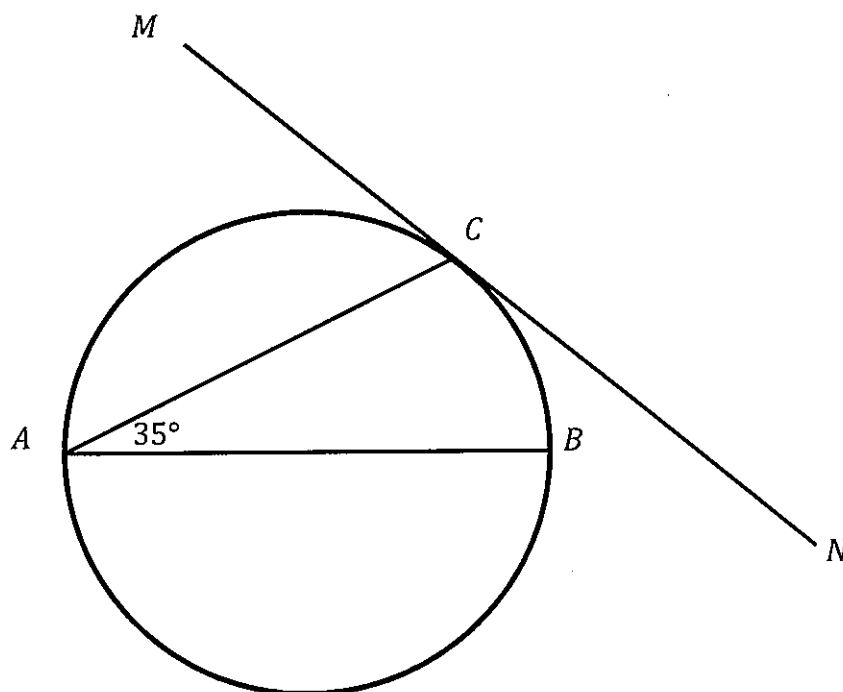
Attempt Questions 1 – 10

Allow about 15 minutes for this section

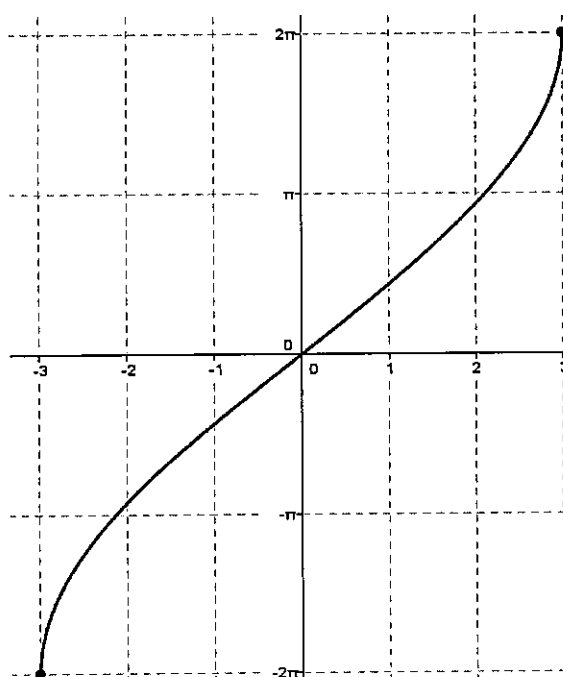
Use the multiple-choice answer sheet for Questions 1–10.

1. In the diagram, AB is a diameter of the circle and MN is tangent to the circle at C . $\angle CAB = 35^\circ$. What is the size of $\angle MCA$?

- (A) 35°
- (B) 45°
- (C) 55°
- (D) 65°



2. Which function is graphed below?



- (A) $2\pi \sin 3x$
- (B) $2\pi \sin^{-1} \frac{1}{3}x$
- (C) $4 \sin^{-1} 3x$
- (D) $4 \sin^{-1} \frac{1}{3}x$

Section I (cont'd)

3. Find $f^{-1}(x)$, given $f(x) = \frac{3x-3}{x-2}$

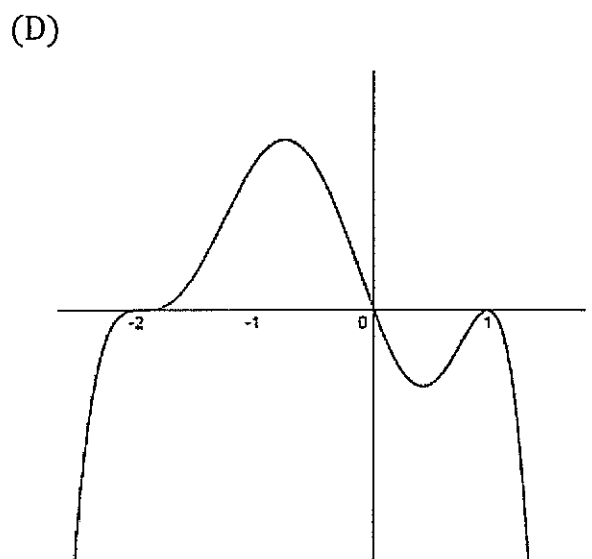
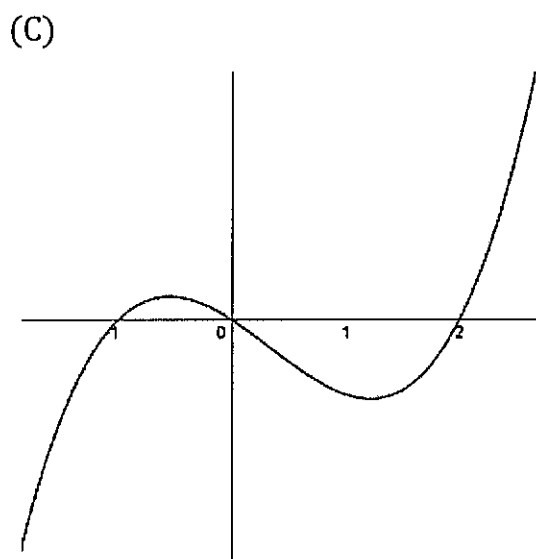
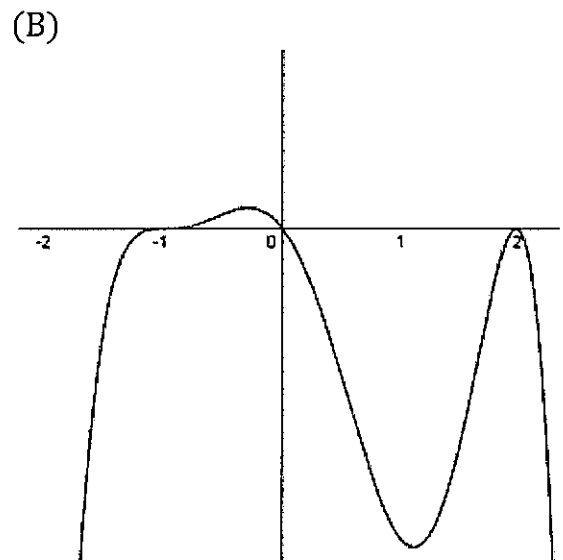
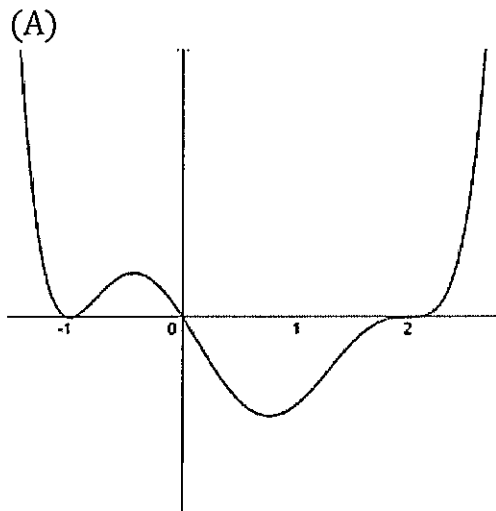
(A) $f^{-1}(x) = \frac{3y-3}{x-2}$

(B) $f^{-1}(x) = \frac{2x-3}{x-3}$

(C) $f^{-1}(x) = \frac{x-2}{3x-3}$

(D) $f^{-1}(x) = \frac{3-3x}{2-x}$

4. Which diagram best represents $y = -x(2-x)^3(x+1)^2$?



Section I (cont'd)

5. Find k given $x - 2$ is a factor of $P(x) = x^3 - 3x^2 + kx + 12$
- (A) $k = -4$
(B) $k = 0$
(C) $k = 2$
(D) $k = 4$
6. The acute angle between $l_1: 2x - y - 3$ and $l_2: y = 3x + 7$ is closest to:
- (A) 15°
(B) 8°
(C) 82°
(D) 45°
7. $\int 2\cos^2 x \, dx$
- (A) $\sin x \cos x + x + C$
(B) $-\frac{1}{2} \sin 2x + x + C$
(C) $\frac{2}{3} \cos^3 x + C$
(D) $\frac{-2}{\sqrt{1-x^2}} + C$
8. The velocity of a particle at a position x is $\dot{x} = 2e^{-\frac{x}{2}}$ metres per second.
Calculate the particle's acceleration when its displacement is -2 metres.
- (A) $-e \text{ m/s}^2$
(B) $-\frac{4}{e^2} \text{ m/s}^2$
(C) $-2e^2 \text{ m/s}^2$
(D) $e^2 \text{ m/s}^2$

Section I (cont'd)

9. Find the exact value of $\sin 15^\circ$

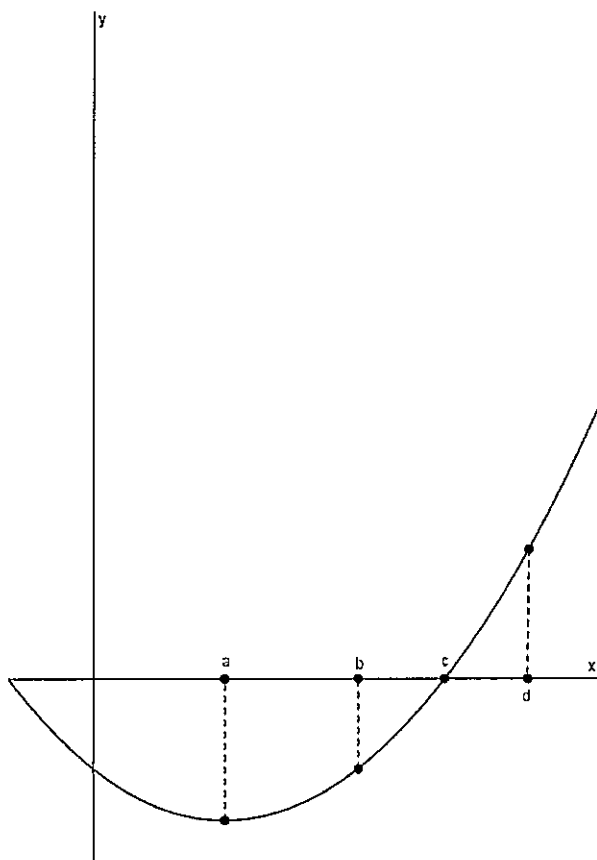
(A) $\frac{1}{3\sqrt{2}}$

(B) $\frac{2-\sqrt{2}}{2\sqrt{2}}$

(C) $2(\sqrt{6} - \sqrt{2})$

(D) $\frac{\sqrt{6} - \sqrt{2}}{4}$

10. Given the curve below, Eden intends to use Newtons Method to find an approximation to the root shown. Which initial estimate will not produce a good approximation with this method?



(A) $x_0 = a$

(B) $x_0 = b$

(C) $x_0 = c$

(D) $x_0 = d$

End of Section I

Section II

60 marks

Attempt Questions 11 – 14

Allow about 1 hours 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11 – 14, your responses should include relevant mathematical reasoning and/or calculations.

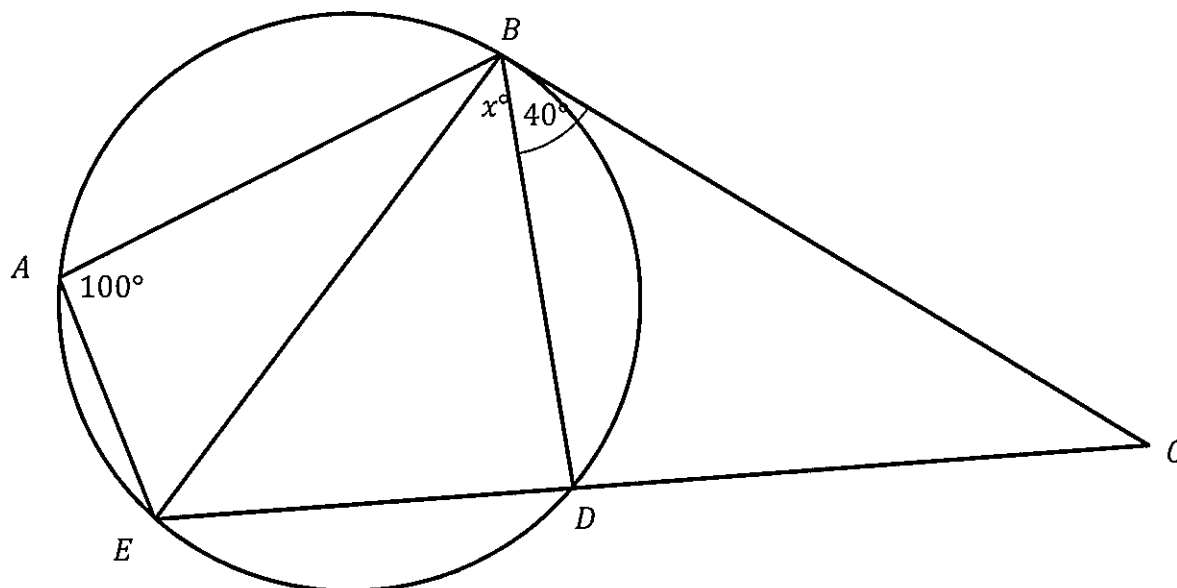
Question 11 (15 marks) Use a SEPARATE writing booklet	Marks
a) Solve the inequality $\frac{1}{ x-1 } > \frac{1}{2}$	2
b) Sketch the intersection of $y \geq x - 1$ and $y < 1$	3
c) Given $A(-2, 3)$ and $B(10, 11)$, find the coordinates of the point P which divides the interval AB in the ratio 3:1.	2
d) You are given 3.6 as an approximate root of the equation $x^3 - 50$. Use one application of Newton's method to find a better approximation. (to 2 decimal places)	2
e) If $y = \sin(\ln x)$, find	
(i) $\frac{dy}{dx}$	(ii) $\frac{d^2y}{dx^2}$ 1,2

Question 11 (continued)

Marks

- f) BC is tangent to the circle at B . Find the value of x , giving reasons.

3



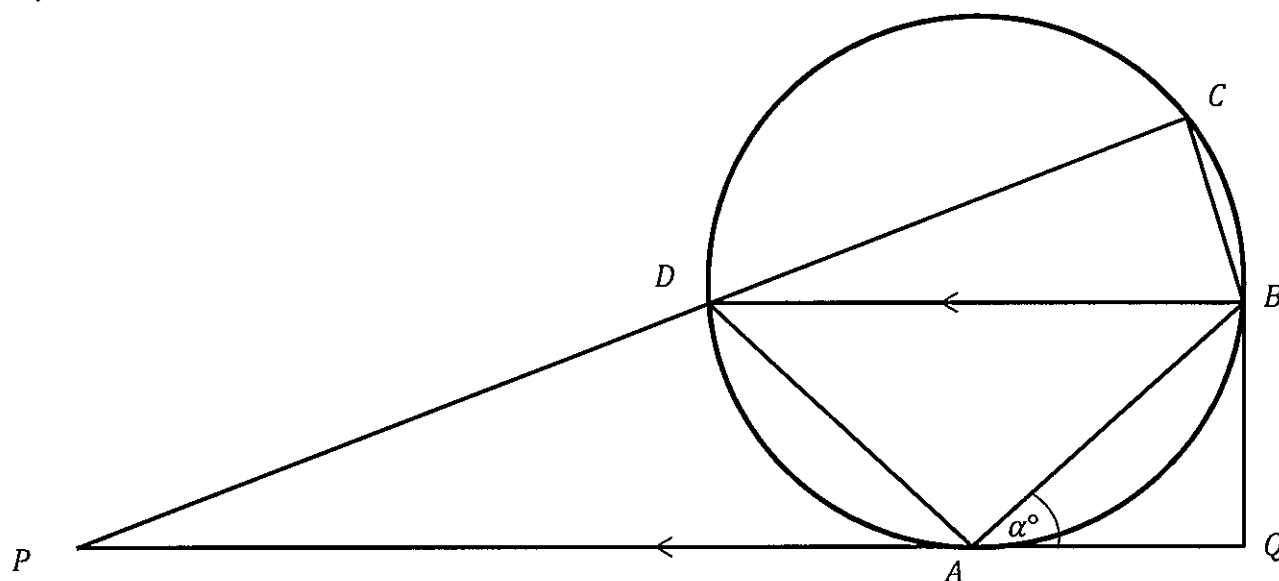
Question 12 (15 marks) Use a SEPARATE writing booklet

Marks

a) Solve $3 \sin x + 4 \cos x = 2.5$, $0 \leq x \leq 2\pi$

3

b)



The tangents from Q touch the circle at A and B . PC and PQ are straight lines $\angle BAQ = \alpha$.

(i) Copy or trace the diagram into your writing booklet.

1

(ii) Given $PD = 5$ cm and $DC = 7$ cm, calculate the exact length of AP .

1

(iii) Show that $\angle BCD = 2\alpha$.

3

(iv) Show that $PQBC$ is a cyclic quadrilateral.

2

Question 12 (continued)

Marks

- c) The rate at which a body warms in air is proportional to the difference between its temperature T and the constant temperature A of the surrounding air. This rate can be expressed by the differential equation

$$\frac{dT}{dt} = k(T - A)$$

where t is time in minutes and k is constant.

- (i) Show that $T = A + Ce^{kt}$ where C is a constant, is a solution of the differential equation. 1
- (ii) A glass of milk warms from 4°C to 8°C in 15 minutes. The air temperature is 25°C . Find the temperature of the glass of milk after a further 45 minutes, correct to the nearest degree. 3
- (iii) With reference to the equation for T , explain the behaviour of T as t becomes very large. 1

Question 13 (15 marks) Use a SEPARATE writing booklet

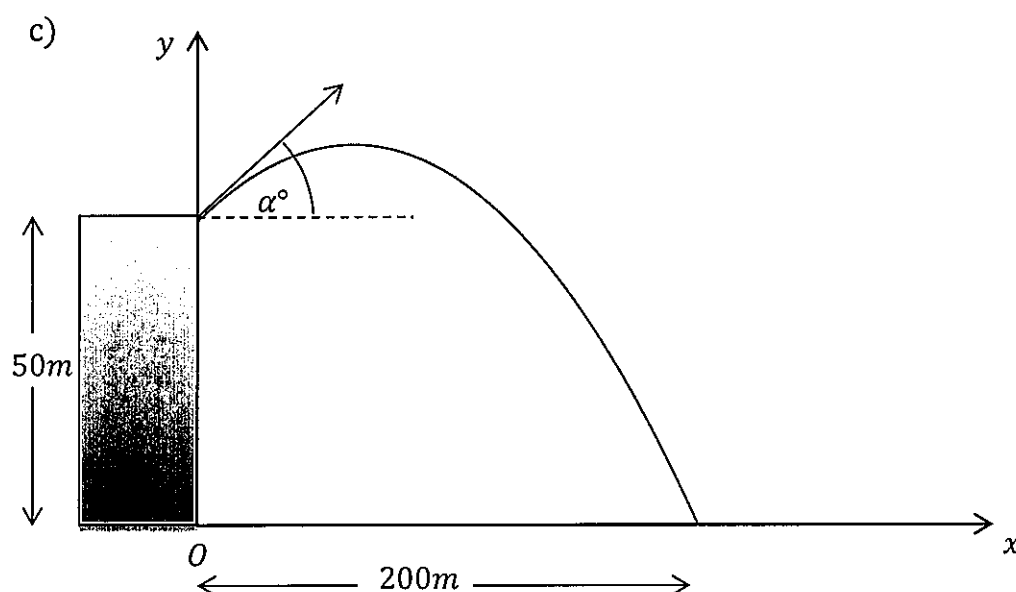
Marks

a) Evaluate $\int_0^1 x^3 (\sqrt{x^4 + 1}) dx$ using the substitution $u = x^4 + 1$. 3

b) (i) By considering its second derivative, show that $y = e^x - 4x$ is always concave up. 2

(ii) Use the trapezoidal rule with 3 function values to find an approximation to $\int_3^5 (e^x - 4x) dx$, correct to 4 significant figures. 3

(iii) Is this approximation too large or too small? Justify your answer? 1



A projectile is launched from the top of a 50 m high building with an initial speed of 40 m/s. It is launched at an angle of α° above the horizontal, as in the diagram. Acceleration due to gravity is 10 m/s^2 .

(i) Given that $\frac{d^2x}{dt^2} = 0$ and $\frac{d^2y}{dt^2} = -10$, show that $x = 40t \cos \alpha$ and $y = -5t^2 + 40t \sin \alpha + 50$ where x and y are the horizontal and vertical displacements of the projectile in metres from 0 at time t seconds after launching. 3

(ii) The projectile lands on the ground 200 metres from the base of the building. Find the two possible values of α . Give your answers to the nearest degree. 3

Question 14 (15 marks) Use a SEPARATE writing booklet **Marks**

- a) $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are points on the parabola $x^2 = 4ay$.
If $p + q = 4$, find the locus of M , the mid point of PQ . 3

- b) Given that x is a positive integer, prove by the method of mathematical induction that $(1 + x)^n - 1$ is divisible by x for all positive integers $n \geq 1$. 3

- c) The velocity $v \text{ ms}^{-1}$ of a particle moving on a horizontal line is given by

$$v^2 = 252 + 216x - 36x^2$$

- (i) Show that the particle is performing simple harmonic motion. 1

- (ii) Find the centre of the motion. 1

- (ii) Find the amplitude of the motion. 1

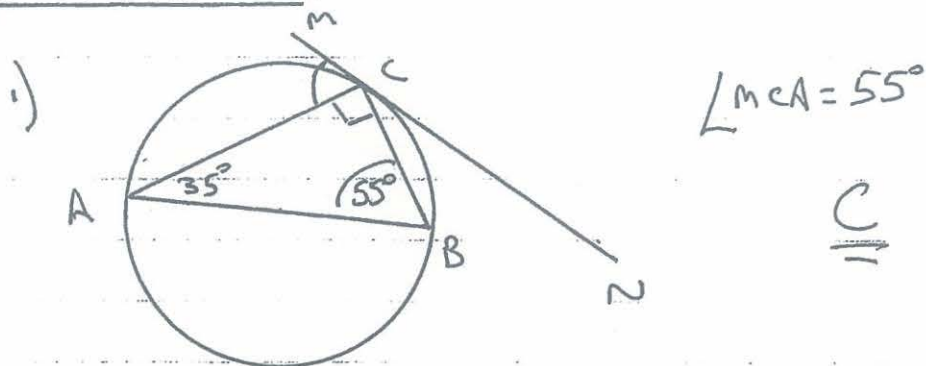
- (iv) Find the period of the motion. 1

- (v) Find the maximum speed of the particle. 1

- (vi) Initially the particle is at one of the extreme points of the motion.
Where will it be when $t = \frac{13\pi}{12}$ seconds. 2

- (vii) Find its average speed during the first $\frac{13\pi}{12}$ seconds. 2

SECTION 1



2) $y = \sin^{-1} x$

$$-1 \leq x \leq 1$$

$$-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$$

For given graph.

$$-3 \leq x \leq 3$$

$$-2\pi \leq y \leq +2\pi$$

$$\text{So } -1 \leq \frac{x}{3} \leq 1$$

$$-\frac{\pi}{2} \leq \frac{y}{4} \leq \frac{\pi}{2}$$

$$\frac{y}{4} = \sin^{-1} \frac{x}{3}$$

$\therefore$ D

3) If $y = \frac{3x-3}{x-2}$

inverse function is $x = \frac{3y-3}{y-2}$

$$xy - 2x = 3y - 3$$

$$y(x-3) = 2x-3$$

$$y = \frac{2x-3}{x-3}$$

So for $f(x) = \frac{3x-3}{x-2}$

$$f^{-1}(x) = \frac{2x-3}{x-3}$$

$\therefore$ B

TIONS

$$4) y = -x(2-x)^3(x+1)^2$$

Single zero at $x=0$ ALL
 Double zero at $x=-1$ A,
 Triple zero at $x=2$ A,

$\therefore \underline{\underline{A}}$

$$5) P(2) = 0$$

$$2^3 - 3(2)^2 + k(2) + 12 = 0$$

$$8 - 12 + 2k + 12 = 0$$

$$2k = -8$$

$$k = -4$$

$\underline{\underline{A}}$

$$6) \tan \theta = \frac{m_2 - m_1}{1 + m_1 m_2} \quad \begin{matrix} m_1 = 2 \\ m_2 = 3 \end{matrix}$$

$$= \frac{3 - 2}{1 + 3 \times 2}$$

$$= \frac{1}{7}$$

$\therefore \underline{\underline{B}}$

$$7) 2 \int \cos^2 x \, dx$$

$$= 2 \int \left[\frac{1}{2} (1 + \cos 2x) \right] dx$$

$$= \int (1 + \cos 2x) dx$$

$$= x + \frac{1}{2} \sin 2x + C$$

$$= x + \sin x \cos x + C$$

$\underline{\underline{A}}$

$$8) \dot{x} = 2e^{-x/2}$$

$$\ddot{x} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

$$= \frac{d}{dx} \left(\frac{1}{2} \cdot 4e^{-x} \right)$$

$$= -2e^{-x}$$

at $x = -2$ $\ddot{x} = -2e^2 \therefore C$

$$9) \sin 15^\circ = \sin (45^\circ - 30^\circ)$$

$$= \sin 45^\circ \cos 30^\circ - \sin 30^\circ \cos 45^\circ$$

$$= \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} - \frac{1}{2} \times \frac{1}{\sqrt{2}}$$

$$= \frac{\sqrt{3} - 1}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$$

$$= \frac{\sqrt{6} - \sqrt{2}}{4} \therefore D$$

10) $x_0 = a$ as tangent nowhere
near $x = c$

A

Question 11

X1 trial 2015 Question 11 (15 marks)

Comments

Method 1

a) $\frac{1}{|x-1|} > \frac{1}{2}$

NB $x \neq 1$

Method 1: $x \neq 1$

Since $|x-1|$ is positive

$$2 > |x-1|$$

$$|x-1| < 2$$

$$-2 < x-1 < 2 \quad x \neq 1$$

$$-1 < x < 3 \quad x \neq 1$$

OR

Method 2

$$x \neq 1$$

if $x-1$ is negative
 $x < 1$

$$\frac{1}{|x-1|} > \frac{1}{2}$$

$$-\frac{1}{(x-1)} > \frac{1}{2}$$

$$\frac{1}{1-x} > \frac{1}{2}$$

$$2 > 1-x \quad (1-x > 0)$$

$$x > -1$$

$$\therefore x > -1 \text{ and } x < 1$$

$$\therefore -1 < x < 1$$

Combining get

$$-1 < x < 1 \quad \text{or} \quad 1 < x < 3$$

OR

$$-1 < x < 3 \quad x \neq 1$$

Method 3 $|x-1| < 2 \quad x \neq 1$ taking reciprocals

$$-2 < x-1 < 2$$

$$-1 < x < 3 \quad x \neq 1$$

Method 4: Graphical method - mainly extension 2 students.

Method 1

Students using this method were generally successful, although some lost marks for not stating $x \neq 1$.

Method 2 (Cases)

Students tried to multiply by $(x-1)^2$ on both sides

if $x-1$ is positive as we generally do with inequalities of the form

$$\frac{1}{|x-1|} > \frac{1}{2}$$

$$\frac{1}{x-1} > \frac{1}{2}$$

$$2 > x-1$$

$$3 > x$$

$$x < 3$$

$$\therefore x > 1 \text{ and } x < 3$$

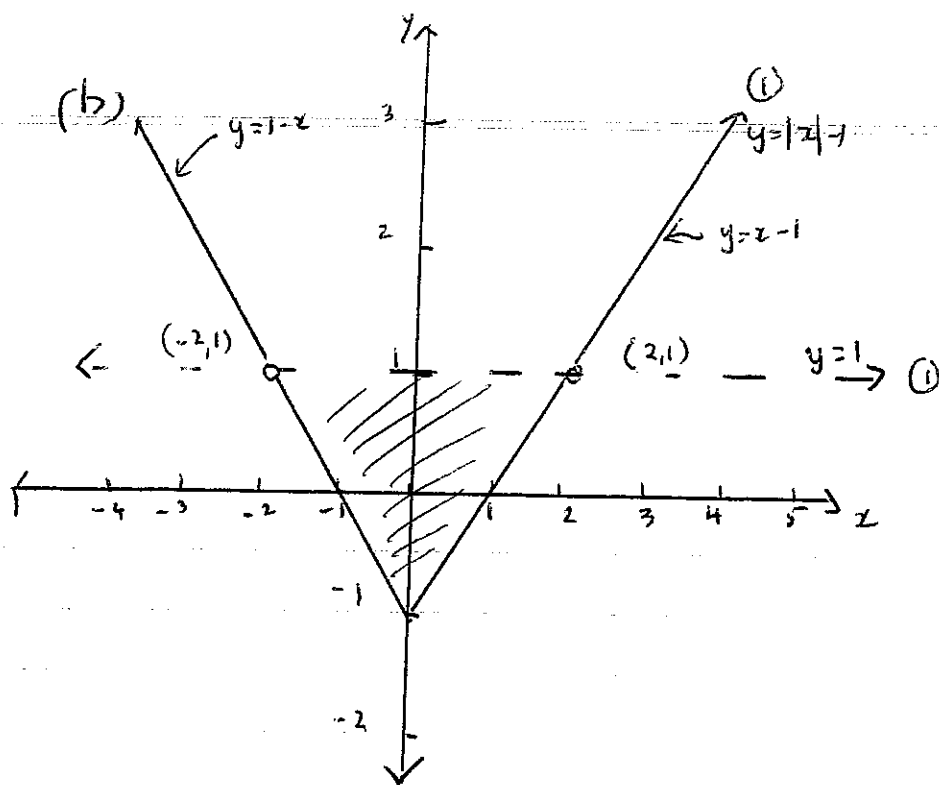
$$\therefore 1 < x < 3$$

$$\frac{1}{|x-1|} > \frac{1}{2}$$

Many found it difficult dealing with $\frac{(x-1)^2}{|x-1|}$ and looking at cases although some did this successfully.

Method 3

if taking reciprocals, if $a > b > 0$ then $\frac{1}{a} < \frac{1}{b}$
NB inequalities are reversed.



3 marks

generally well done
many students
did NOT clearly
mark the points
 $(-2, 1)$ and $(2, 1)$
with open circles
and lost marks $(\frac{1}{2})$

(c) Divide AB in the ratio 3:1

$$A(-2, 3) \quad B(10, 11)$$

3 : 1

$$\text{Point} = \left(\frac{3(10) + 1(-2)}{3+1}, \frac{3(11) + 1(3)}{3+1} \right)$$

①mk

$$= \left(\frac{30-2}{4}, \frac{33+3}{4} \right)$$

$$= (7, 9)$$

①mk.

2 marks.

(d) $x_0 = 3.6$

$$f(x) = x^3 - 50$$

$$f(3.6) = 3.6^3 - 50$$

$$f'(x) = 3x^2$$

$$f'(3.6) = 3(3.6)^2$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$= 3.6 - \frac{3.6^3 - 50}{3(3.6)^2} \quad (1)$$

$$= 3.69 \text{ (2 dp)}$$

generally well done
some forgot the
formula - many

reversed the
numerator & denominator
in the formula.

Question 13

Solution

$$a) \int_0^1 x^3 \sqrt{x^4+1} dx$$

$$\text{Let } u = x^4 + 1$$

$$du = 4x^3 dx$$

$$\frac{1}{4} du = x^3 dx$$

$$\text{when } x=1, u=2$$

$$x=0, u=1$$

$$\int_1^2 x^3 \sqrt{x^4+1} dx = \int_1^2 \sqrt{u} \cdot \frac{1}{4} du$$

$$= \frac{1}{4} \int_1^2 u^{1/2} du$$

$$= \frac{1}{4} \left[\frac{2}{3} u^{3/2} \right]_1^2$$

$$= \frac{1}{6} \left[2^{3/2} - 1 \right]$$

$$= \frac{1}{6} (2\sqrt{2} - 1)$$

$$= \frac{\sqrt{2}}{3} - \frac{1}{6}$$

Comment

• 1 mark for correct substitution and change of limits

• 1 mark - correct integration

} • 1 mark for answer

3

Q13

b) i) $y = e^x - 4x$

$$y' = e^x - 4$$

$$y'' = e^x$$

as $e^x > 0$ for all values of x

$\therefore y'' > 0$ for all x

So the curve is always concave up.

ii) $h = \frac{b-a}{n}$
 $= \frac{5-3}{2}$
 $= 1$

$$\begin{array}{ccc} 3 & 4 & 5 \\ | & | & | \\ y_0 & y_1 & y_2 \\ (e^3-12) & (e^4-16) & (e^5-20) \end{array}$$

$$\begin{aligned} \int_3^5 e^x - 4x \, dx &\doteq \frac{1}{2} \{ y_0 + 2 \times y_1 + y_2 \} \\ &= \frac{1}{2} \{ e^3 - 12 + 2(e^4 - 16) + e^5 - 20 \} \\ &= \frac{1}{2} \{ 8.0855 + 2(38.598) + 128.413 \} \\ &= 106.8 \quad (\text{to 4 sig figs}) \end{aligned}$$

Method 2

$$\int_3^5 (e^x - 4x) \, dx$$

$$\doteq \frac{4-3}{2} (e^4 - 16 + e^3 - 12) + \frac{5-4}{2} (e^5 - 20 + e^4 - 16)$$

$$\doteq \frac{1}{2} (e^4 + e^3 - 28) + \frac{1}{2} (e^5 + e^4 - 36)$$

$$\doteq 106.8$$

This was answered well.

• 1 mark for derivatives

• 1 mark for stating $e^x > 0$ and $y'' > 0$

(2)

• 1 mark for finding h correctly

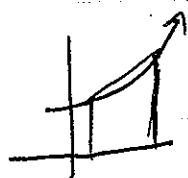
} 1 mark for correctly using trapezoidal rule

• 1 mark for answer

(3)

Q13

b) iii)



It is too large as the trapezium area is larger than the area under the curve as the function concaves up for all x and $f(3) < f(5)$ so the function has



shape

1 mark

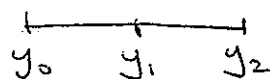
Note: For those students who only said it was too large without a reason, I awarded only $\frac{1}{2}$ mark.

Examiners comment for

13 b ii) The approximation ~~can be~~ found two ways:

Method 1 using the formula for the trapezoidal rule for n -equal subintervals

$$\frac{h}{2} \{y_0 + 2y_1 + y_2\}$$



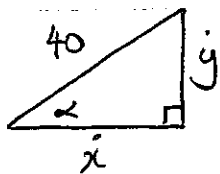
and

Method 2 : Using one application of trapezoidal rule twice.

Some Students who used Method 1, did not find h ~~correct~~ correctly and confused the rule by multiply y_2 by 2 instead of y_1 .

Q13

c) i) Initial conditions, $t=0$



$$v_x \quad \dot{x} = 40 \cos \alpha$$

$$v_y \quad \dot{y} = 40 \sin \alpha$$

Horizontal component

$$\ddot{x} = 0$$

$$\dot{x} = c$$

$$\text{when } t=0, \dot{x} = 40 \cos \alpha$$

$$\therefore c = 40 \cos \alpha$$

$$\therefore \dot{x} = 40 \cos \alpha$$

$$x = 40t \cos \alpha + c$$

$$\text{when } t=0, x=0,$$

$$\therefore c=0$$

$$\therefore x = 40t \cos \alpha \quad \text{--- (1)}$$

Vertical

$$\ddot{y} = -10$$

$$\dot{y} = -10t + c$$

$$\text{when } t=0, \dot{y} = 40 \sin \alpha$$

$$\therefore c = 40 \sin \alpha$$

$$\therefore \dot{y} = -10t + 40 \sin \alpha$$

$$y = -5t^2 + 40t \sin \alpha + c$$

$$\text{when } t=0, y=50$$

$$50 = c$$

$$\therefore y = -5t^2 + 40t \sin \alpha + 50 \quad \text{--- (2)}$$

1 for Initial Conditions

Students were penalised for not finding c.

1 for $\dot{x}$ & $\dot{y}$

1 to find c for x and y.

3

ii) when $x=200, y=0$

$$\text{Using } x = 40t \cos \alpha$$

$$200 = 40t \cos \alpha$$

$$t = \frac{5}{\cos \alpha}$$

sub in t and $y=0$ in (2)

$$0 = -5\left(\frac{5}{\cos \alpha}\right)^2 + 40\left(\frac{5}{\cos \alpha}\right) \sin \alpha + 50$$

$$= -125 \sec^2 \alpha + 200 \tan \alpha + 50$$

$$= -125(1 + \tan^2 \alpha) + 200 \tan \alpha + 50$$

$$= -125 - 125 \tan^2 \alpha + 200 \tan \alpha + 50$$

$$= 5 \tan^2 \alpha - 8 \tan \alpha + 3$$

$$= (5 \tan \alpha - 3)(\tan \alpha - 1)$$

$$\tan \alpha = \frac{3}{5} \quad \text{or} \quad \tan \alpha = 1$$

$$\alpha = 31^\circ (\text{nearest degree}) \quad \alpha = 45^\circ$$

1 sub $x=200$ in (1)

1 sub. $y=0$ and t in (2)

1 for correct angles (3)

Alternative solution to

c) ii)

From (1)

$$t = \frac{x}{40 \cos \alpha}$$

Sub in (2)

$$y = -5 \left(\frac{x}{40 \cos \alpha} \right)^2 + 40 \left(\frac{x}{40 \cos \alpha} \right) \sin \alpha + 50$$

$$= -\frac{5x^2}{1600 \cos^2 \alpha} + x \tan \alpha + 50$$

$$= -\frac{x^2 \sec^2 \alpha}{320} + x \tan \alpha + 50$$

$$= -\frac{x^2 (1 + \tan^2 \alpha)}{320} + x \tan \alpha + 50$$

When $y = 0$

$$0 = -x^2 - x^2 \tan^2 \alpha + 320x \tan \alpha + 16000$$

$$0 = x^2 + x^2 \tan^2 \alpha - 320x \tan \alpha - 16000$$

$$0 = x^2 \tan^2 \alpha - 320x \tan \alpha + x^2 - 16000$$

but $x = 200$

$$0 = 40000 \tan^2 \alpha - 64000 \tan \alpha + 24000$$
$$= 5 \tan^2 \alpha - 8 \tan \alpha - 3$$

Then as per method 1

This method was not the optimal method. It allowed for more errors.

1 mark was awarded here, when

$$t = \frac{x}{40 \cos \alpha} \text{ was}$$

substituted in (1)

1 mark was awarded when $y = 0, x = 200$ was subst.

(No marks were awarded when students took $y = -50$).

1 mark for correct angles.

QUESTION 12

SOLUTION

(a) Let $R \sin(x+\alpha) = 3 \sin x + 4 \cos x$ — ①

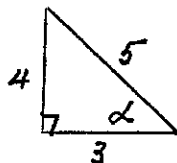
$$\therefore R \sin x \cos \alpha + R \cos x \sin \alpha$$

$$= 3 \sin x + 4 \cos x.$$

$$\therefore \begin{cases} R \cos \alpha = 3 \\ R \sin \alpha = 4 \end{cases} \Rightarrow \alpha \text{ is acute}$$

$$\therefore \tan \alpha = \frac{4}{3}$$

$$\alpha = \tan^{-1}\left(\frac{4}{3}\right)$$



$$\therefore R \times \frac{3}{5} = 3$$

$$R = 5$$

$$\text{①} \Rightarrow 5 \sin(x+\alpha) = 2.5$$

$$\sin(x+\alpha) = 0.5$$

$$\therefore x+\alpha = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}, \dots$$

$$\therefore x = \frac{\pi}{6} - \tan^{-1}\left(\frac{4}{3}\right), \frac{5\pi}{6} - \tan^{-1}\left(\frac{4}{3}\right)$$

$$\frac{13\pi}{6} - \tan^{-1}\frac{4}{3}, \dots$$

$$= -0.41, 1.69, 5.88, \dots$$

(correct to 2 dec. places)

BUT $0 \leq x \leq 2\pi \Rightarrow x = 1.69, 5.88.$

OR Using t -substitution

$$3 \sin x + 4 \cos x = 2.5$$

$$\Rightarrow 3 \cdot \frac{2t}{1+t^2} + 4 \cdot \frac{1-t^2}{1+t^2} = 2.5$$

$$\Rightarrow 6t + 4 - 4t^2 = 2.5 + 2.5t^2$$

$$0 = 6.5t^2 - 6t - 1.5$$

$$\text{ie } 13t^2 - 12t - 3 = 0$$

$$\therefore t = \frac{12 \pm \sqrt{144 + 156}}{26}$$

$$= \frac{12 \pm 10\sqrt{3}}{26}$$

$$= \frac{6 \pm 5\sqrt{3}}{13}$$

COMMENTS

• Some students squared both sides but did not check their solutions.

• 1 MARK for $5 \sin(x + \tan^{-1}\frac{4}{3}) = 2.5$

• Some students made no mention of radians.

• Many students missed the solution $x = \frac{13\pi}{6} - \tan^{-1}\left(\frac{4}{3}\right)$

and did not realise that $x = \frac{\pi}{6} - \tan^{-1}\left(\frac{4}{3}\right)$

was outside the domain. (2 MARKS)

SOLUTION

COMMENTS

$$\therefore \tan \frac{x}{2} = 1.1277, -0.2046$$

$$\therefore x = 1.69, 5.88 \text{ (correct to 2 dec. pl.)}$$

- Need to check $x = \pi$ when using this method.

(b) (i)

$$(ii) AP^2 = PC \cdot PD$$

$$= 12 \times 5$$

$$= 60$$

$$AP > 0 \Rightarrow AP = \sqrt{60}$$

$$= 2\sqrt{15}$$

- All students were successful!
- Too many students wrote $AP^2 = 7 \times 5$.

$$(iii) \hat{ABD} = \hat{BAQ} = x \text{ (alternate angles, } DB \parallel AQ)$$

$$\hat{ADB} = \hat{BAQ} = x \text{ (angle between tangent and chord equals angle in alternate segment)}$$

$$\therefore \hat{BAD} = 180^\circ - 2x \text{ (angle sum of triangle ABD)}$$

$$\hat{BAD} + \hat{BCD} = 180^\circ \text{ (opposite L's of cyclic quad are supplementary)}$$

$$\therefore 180 - 2x + \hat{BCD} = 180^\circ$$

$$\therefore \hat{BCD} = 2x$$

- Generally gained at least 1 mark for correct statement and reason.

$$(iv) QA = QB \text{ (tangents to circle from external point are equal)}$$

$$\therefore \hat{QBA} = \hat{QAB} = x \text{ (equal angles opposite equal sides in isosceles triangle)}$$

$$\therefore \hat{AQB} = 180^\circ - 2x \text{ (angle sum of triangle is } 180^\circ)$$

$$\hat{AQB} + \hat{BCD} = 180^\circ - 2x + 2x$$

$$= 180^\circ$$

$\therefore PQBC$ is cyclic quad. as opposite angles are supplementary

- Generally lose 1 mark for any missing reason.

$$(c) \frac{dT}{dt} = k(T-A)$$

$$(i) \text{ If } T = A + Ce^{kt} \quad \text{--- (1)}$$

$$\begin{aligned} \text{Then } \frac{dT}{dt} &= Cke^{kt} \\ &= k \cdot Ce^{kt} \\ &= k(T-A) \text{ from (1)} \end{aligned}$$

- Generally well done - no need to DERIVE the result by integration.

$$(ii) \quad T = A + Ce^{kt} \quad A = 25$$

$$\text{at } t=0, T=4$$

$$\therefore 4 = 25 + Ce^0$$

$$\therefore C = -21$$

$$\therefore T = 25 - 21e^{kt}$$

$$\left. \begin{array}{l} t=15 \\ T=8 \end{array} \right\} \Rightarrow 8 = 25 - 21e^{15k}$$

$$\Rightarrow e^{15k} = \frac{17}{21}$$

$$\therefore k = \frac{\ln\left(\frac{17}{21}\right)}{15} \quad \text{--- (2)}$$

$$\begin{aligned} \text{at } t=60 \quad T &= 25 - 21e^{60k} \\ &= 15.98^\circ\text{C} \text{ (2 dec. pl)} \\ &= 16^\circ\text{C to nearest degree} \end{aligned}$$

- Some students were unable to begin this question, others were confused, thought that $t=15$ when $T=4$

- Find C correctly
1 MARK

- Find k correctly
1 MARK

- Use of $t=45$ scored no mark.

$$(iii) \quad T = 25 - 21e^{kt}$$

$$\text{From (2) } k < 0$$

$$\text{Hence as } t \rightarrow \infty \quad T \rightarrow 25$$

$$\text{as } e^{kt} \rightarrow 0$$

- Some students thought that $T \rightarrow \infty$ as $t \rightarrow \infty$ because they failed to see that $k < 0$.

QUESTION 14

a) $M = \left(\frac{2ap+2aq}{2}, \frac{ap^2+aq^2}{2} \right)$ 1 mark

$$\begin{aligned}\therefore x &= \frac{2ap+2aq}{2} \\ &= a(p+q) \\ &= 4a\end{aligned}$$

$\therefore$ The locus of M is
the vertical line $x=4a$

b) Prove true for $n=1$

When $n=1$,

$$\begin{aligned}(1+x)^1 - 1 &= 1+x-1 \\ &= x\end{aligned}$$

which is divisible
by x

Assume true for $n=k$

That is,

$$(1+x)^k - 1 = Mx,$$

where M is an
integer

1 mark for correctly
using $p+q=4$

1 mark for concluding
that the locus is
 $x=4a$. Note that
this had to be your
answer, not just
floating around in your
working.

1 mark for correctly
proving true for $n=1$
and making the
assumption

Prove true for $n = k+1$

That is,

$(1+x)^{k+1} - 1$ is divisible by x

$$(1+x)^{k+1} - 1 = (1+x)^k (1+x) - 1$$

$$= (1+x)^k + x(1+x)^k - 1$$

$$= x(1+x)^k + (1+x)^k - 1$$

$$= x(1+x)^k + Mx$$

by assumption

$$= x((1+x)^k + M)$$

which is divisible by x .

$\therefore$ the statement is true for $n = k+1$ if it is true for $n = k$.

It is true for 1, so it is true for 2, and therefore true for all n .

1 mark for correctly using your assumption.

1 mark for reducing to something obvious, divisible by x , and for concluding with an appropriate statement.

c)

$$v^2 = 252 + 216x - 36x^2$$

$$\frac{1}{2}v^2 = 126 + 108x - 18x^2$$

$$\begin{aligned}\ddot{x} &= \frac{d}{dx} \left(\frac{1}{2}v^2 \right) \\ &= \frac{d}{dx} (126 + 108x - 18x^2)\end{aligned}$$

$$= 108 - 36x$$

$$= -36(x-3)$$

which is of the form

$$\ddot{x} = -n^2(x-x_0)$$

$\therefore$ the particle is in simple harmonic motion

OR

$$v^2 = 252 + 216x - 36x^2$$

$$= 36(7 + 6x - x^2)$$

$$= 36(7+9 - 9 + 6x - x^2)$$

$$= 36[16 - (x^2 - 6x + 9)]$$

$$= 36[16 - (x-3)^2]$$

which is of the form

$$n^2(a^2 - (x-x_0)^2)$$

$\therefore$ simple harmonic motion

This is a "show" question, and there is very little flexibility in the form your answer must take

[36(3-x) is worth half marks]

i) $x = 5m$

1 mark

ii) Extremes when $v=0$

$$36x^2 - 216x - 252 = 0$$

$$x^2 - 6x - 7 = 0$$

$$(x-7)(x+1) = 0$$

$$\therefore x = 7 \text{ or } x = -1$$

$$\text{Amplitude} = \frac{7 - (-1)}{2}$$

$$= 4 \text{ m}$$

1 mark

iv) $n^2 = 36$ so $n = 6$

$$\text{Period} = \frac{2\pi}{n}$$

$$= \frac{2\pi}{6}$$

$$= \frac{\pi}{3} \text{ s}$$

1 mark

v) Maximum speed at centre
ie when $x=3$

$$v^2 = 252 + 648 - 324$$

$$= 576$$

$$v = \pm 24 \text{ but speed} > 0$$

$\therefore$ maximum speed is 24 ms^{-1}

The question asks
for speed, so
don't give $|V_{\max}|$
or similar.

1 mark

vi) Motion of the form

$$x = x_0 + a \cos(nt)$$

$$= 3 + 4 \cos 6t$$

1 mark

when $t = \frac{13}{12}$,

$$x = 3 + 4 \cos\left(6 \times \frac{13\pi}{12}\right)$$

$$= 3 + 4 \cos \frac{13\pi}{2}$$

$$= 3 + 4 \times 0$$

$$= 3 \text{ m}$$

1 mark

VII) As period is $\frac{\pi}{3}$,

$$\frac{13\pi}{12} \div \frac{\pi}{3} = \frac{13}{4}$$

$$= 3\frac{1}{4}$$

The particle has completed

$3\frac{1}{4}$ oscillations

1 oscillation is $4 \times \text{amplitude}$
 $= 16 \text{ m}$

$$\therefore \text{total distance} = 3\frac{1}{4} \times 16$$

$$= 52 \text{ m}$$

1 mark

$$\text{Average speed} = \frac{\text{distance}}{\text{time}}$$

$$= \frac{52}{\frac{13\pi}{12}}$$

$$= \frac{48}{\pi} \text{ ms}^{-1}$$

1 mark

This question requires some understanding of periodicity -

finding average speed without that is worth zero